

# Introduction to AI and Generative AI

What artificial intelligence really is, the different types, how it learns, and where generative AI fits in.

WHAT IS AI

TYPES & MODELS

GEN AI VS PREDICTIVE AI

RISKS & BENEFITS

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# What we will cover

Eight short stops. By the end you should be able to explain AI to someone with no technical background.

01

## What is AI?

A plain-English definition and everyday examples

02

## Types of AI

By capability and by the way it works

03

## How AI models learn

The four learning styles, minus the maths

04

## What is Generative AI?

Machines that create, not just decide

05

## Gen AI vs Predictive AI

The single most useful comparison

06

## Advantages

Why every business is investing in it

07

## Risks and limitations

Where it goes wrong, and why

08

## Using it well

A simple rule for safe, sensible use

SECTION 01

# What Is Artificial Intelligence?

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Starting with the basics – in normal language.

# Artificial Intelligence

Software that learns from data to do jobs that used to need a human brain.

- It looks at large amounts of past data
- It finds patterns in that data
- It uses those patterns on something new
- It gets better as it sees more examples
- No one writes step-by-step rules for it

## ■ Traditional software

A person writes the rules. Same input always gives the same output.

## ■ Artificial intelligence

The machine works out the rules from examples it has been shown.

## ■ Why it matters

You can now automate judgement, not just repetitive clicking.

# You already use AI several times a day

AI is not new or futuristic. It is already normal.

## ■ Maps and navigation

Predicts traffic and picks the fastest route using live and historic data.

## ■ Streaming and shopping

Suggests the next show or product based on what people like you chose.

## ■ Bank fraud alerts

Spots a payment that does not match your normal spending and blocks it.

## ■ Email spam filter

Learns from millions of flagged emails what junk looks like.

## ■ Face unlock

Matches the camera image to the stored pattern of your face.

## ■ Chat assistants

Understands your question in plain language and writes an answer.

SECTION 02

# Types of AI

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Two useful ways to sort it.

# How smart is it?

Almost every product marketed as 'AI' today is narrow AI.

I

## Narrow AI

Very good at one job only. A chess engine cannot drive a car.

Status: this is everything we use today.

2

## General AI

Would handle any task a human can, and switch between them.

Status: does not exist yet.

3

## Super AI

Would beat humans at everything, including creativity.

Status: theory and debate only.

# Four families you will hear named

They overlap – deep learning is a branch of machine learning.

## ■ Machine Learning

Learns patterns from structured data.  
Used for scores, forecasts and risk ratings.

## ■ Deep Learning

Uses layered neural networks.  
Handles images, audio and language well.

## ■ Natural Language Processing

Reads and writes human language.  
Powers chatbots and translation.

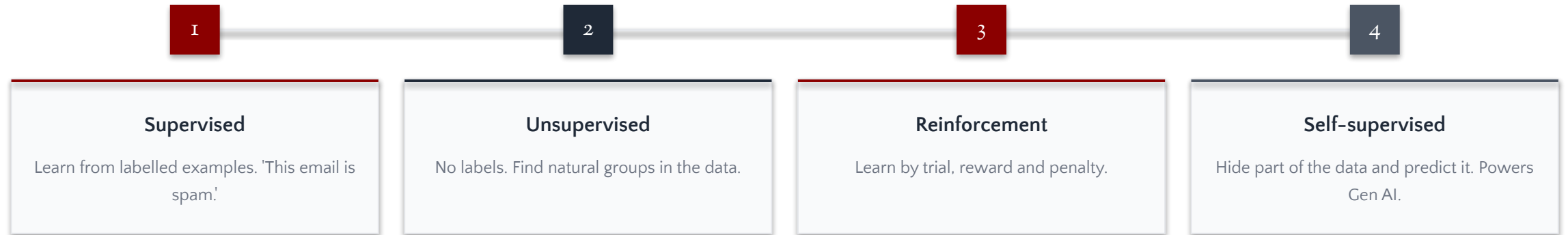
## ■ Computer Vision

Understands pictures and video.  
Used in quality checks and security.



# Four ways to train a model

A 'model' is simply the trained result – the file that holds everything the system learned.



Self-supervised learning is the reason Gen AI arrived so quickly – it can learn from raw internet text without humans labelling it.

SECTION 03

# What Is Generative AI?

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From machines that decide to machines that create.

# Generative AI

AI that creates new content - text, images, audio, video or code - from a simple instruction.

- You give it an instruction, called a prompt
- It predicts the most likely next word, pixel or note
- It repeats that prediction until the output is complete
- The result is new, not copied from a database
- Ask twice and you get two different answers

## ■ Text

Emails, summaries, reports, code - ChatGPT, Claude, Gemini.

## ■ Images and video

Posters, mockups, ads - Midjourney, VEO.

## ■ Audio

Voiceovers, music, dubbing in many languages.

# The main kinds of Gen AI models

You do not need to build these. You need to know which one to pick.

## ■ Large Language Models

Trained on huge amounts of text.  
Understand and write language. GPT,  
Claude, Gemini, Llama.

## ■ Diffusion models

Start from visual noise and clean it up  
step by step into an image.  
Midjourney, Stable Diffusion.

## ■ Multimodal models

Accept and produce more than one  
format – read a chart, answer in  
words.

## ■ Foundation models

Large general models trained once,  
then adapted cheaply to many  
specific tasks.

SECTION 04

# Gen AI vs Predictive AI

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The comparison that clears up most confusion.

# Two different jobs, not two competitors

They work best together, not instead of each other.

## PREDICTIVE AI

Answers: what is likely to happen?

- Output is a number, score or category
- Trained on your own historic data
- Answer is stable and repeatable
- Accuracy can be measured exactly
- Example: which customers will leave next month

## GENERATIVE AI

Answers: can you create this for me?

- Output is new content
- Trained on very large public datasets
- Answer varies each time you ask
- Quality is judged by a human
- Example: write the offer email to keep them

IN SHORT Predictive AI tells you what to expect. Generative AI helps you act on it.

# Quick reference table

|                         | PREDICTIVE AI                 | GENERATIVE AI                      |
|-------------------------|-------------------------------|------------------------------------|
| <b>Main question</b>    | What will happen?             | What can be made?                  |
| <b>Output</b>           | Score, number, label          | Text, image, audio, code           |
| <b>Data used</b>        | Company's own history         | Very large public datasets         |
| <b>Same input twice</b> | Same answer                   | Different answer                   |
| <b>Judged by</b>        | Accuracy against reality      | Human review of quality            |
| <b>Business use</b>     | Forecast demand, score credit | Draft content, summarise, assist   |
| <b>Main weakness</b>    | Fails when the future changes | Can state wrong things confidently |

If you remember one row, remember the first one.

SECTION 05

# Advantages, Risks and Limitations

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An honest look at both sides.



# Why organisations are investing

## ■ Speed

First drafts, summaries and reports in minutes instead of days.

## ■ Lower cost

Routine work is handled by software, so teams handle more with the same headcount.

## ■ Always available

No shifts, no holidays, no drop in quality at 11 pm.

## ■ Consistency

The same standard of output every single time.

## ■ Scale

Serve ten customers or ten thousand with the same system.

## ■ Better decisions

Patterns in large data that a person would simply never spot.

# Risks you must plan for

None of these are reasons to avoid AI. They are reasons to use it with a process.

## ■ Hallucination

- Invents facts, names and figures
- Sounds completely confident while wrong
- Always verify before you use it

## ■ Bias

- Learns the bias in its training data
- Can quietly treat groups unfairly
- A real legal and reputation risk

## ■ Data leakage

- Staff paste confidential data into public tools
- That data may be stored or reused
- Needs a clear company policy

## ■ Over-reliance

- Skills fade when nobody checks the output
- Copy-paste answers with no thought
- Keep a human accountable

# What today's AI genuinely cannot do

Knowing the limits is what separates confident users from careless ones.

- **No real understanding**

It matches patterns in language. It does not know what the words mean.

- **No common sense**

It can pass a professional exam and then fail a question a child would get right.

- **Knowledge has a cut-off date**

Unless it is connected to live search, it does not know recent events.

- **Weak at exact maths and logic**

It predicts likely text, so it can get simple arithmetic wrong.

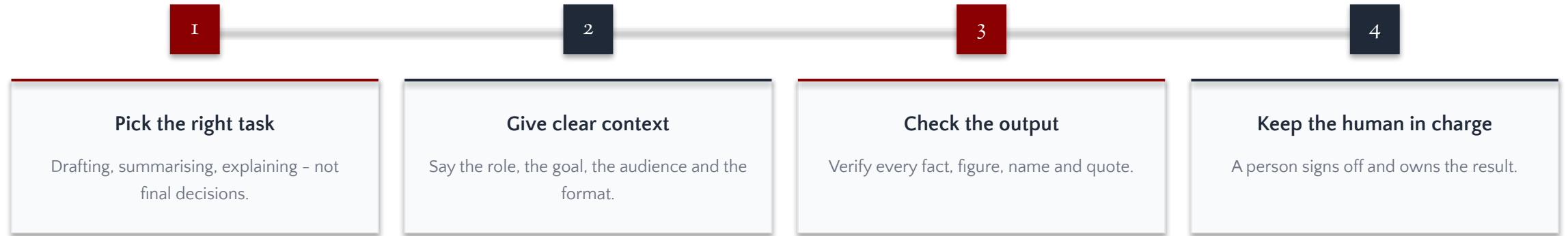
- **Cannot explain itself**

It usually cannot show why it produced a particular answer.

- **No accountability**

A model cannot be responsible. A person always has to be.

# A simple four-step habit



Treat AI as a fast junior colleague: helpful, quick, and always reviewed before anything goes out.

SOMETHING TO REMEMBER

# AI will not replace you. A person using AI well might.

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The skill is no longer doing the task. It is directing the tool and judging the result.

# Key takeaways

Unit 1 in six lines.

1

**AI learns from data instead of following written rules**

That is the one difference that explains everything else.

2

**Everything in use today is narrow AI**

Excellent at one job, useless outside it.

3

**Predictive AI forecasts, Generative AI creates**

Different jobs. Strongest when used together.

4

**The benefits are speed, cost, scale and consistency**

Which is why adoption is moving so fast.

5

**The risks are real but manageable**

Hallucination, bias, data leakage and over-reliance.

6

**A human stays accountable for the output**

Verify, then approve. Always.

CONCLUSION

# Thank you

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Next up — Unit 2: the platforms, tools and prompt-writing skills that turn these ideas into daily work.

QUESTION 1

Which tasks in your own work are drafting rather than deciding?

QUESTION 2

Where would a wrong AI answer cost you the most?

QUESTION 3

What should your team never paste into a public AI tool?